

4 Thermodynamic potentials

We are familiar with the idea of a potential in mechanics, $V(x)$ in one dimension, where x is the position of a particle, which allows the deduction of the force, $F(x)$, on a particle $F = -dV/dx$. Mechanical equilibrium (no net force on the body) occurs at x_0 if $V'(x_0) = 0$ (in simple cases) and the equilibrium is *stable* if $V''(x_0) > 0$.

Thermodynamic potentials partly parallel such a description: they can be used to determine forces on objects in (thermal) contact with a heat bath (for example), due to gradients in entropy as well as gradients of potential and internal energy. We will (probably) not explore those direct parallels in this module (but see Y3 Statistical Physics), although issues of stability are closer.

4.1 The internal energy as a fundamental potential

Let us start by considering the already familiar potential: the internal energy, U , under reversible transformations. Then

$$\begin{aligned} dU &= dW^{\text{rev}} + dQ^{\text{rev}} \\ \text{with } dW^{\text{rev}} &= -p dV \\ \text{and } dS &= \frac{dQ}{T} \Rightarrow dQ = T dS \\ \Rightarrow dU &= T dS - p dV . \end{aligned}$$

Although this was derived for reversible transformations, all functions are functions of state and so this is valid for arbitrary transformations.

This may be generalised to other forms of work (see problem (3.1))

$$dU = T dS - p dV + \sum_i Y_i dX_i , \quad (4.1)$$

where, for example

$$\Gamma d\ell = Y_1 dX_1 \quad \text{and/or} \quad \gamma dA = Y_2 dX_2 ,$$

where Γ is Young's modulus and $d\ell$ is the change in length of an object, or γ is a surface tension and dA is the change in surface area.

Then, comparing the general expression for the total differential for $f(x_1, x_2, x_3, \dots)$,

$$df = \sum_i \frac{\partial f}{\partial x_i} dx_i ,$$

we see that the internal energy is a function $U(S, V, \{X_i\})$, which depends only on extensive parameters (i.e. proportional to the size of the system).

One may see entropy is extensive by combining two bodies each gaining Q , see Fig. (4.1).

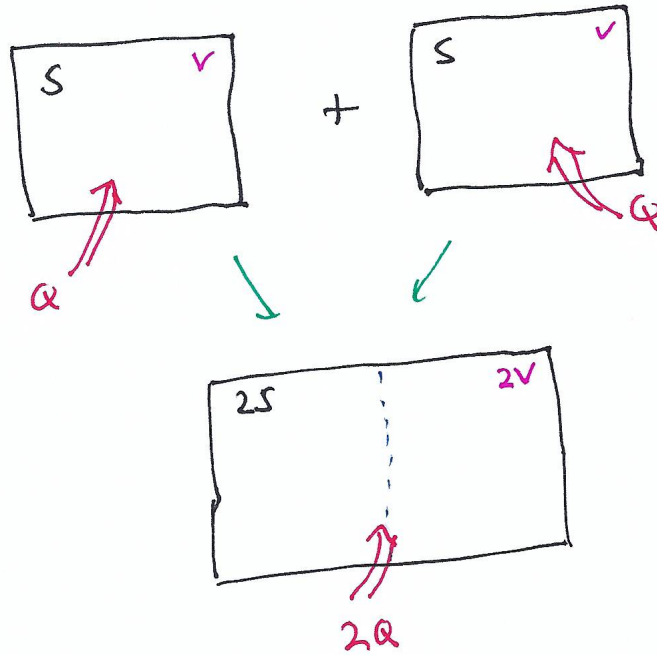


Figure 4.1: The extensivity of entropy, S , shown by combining two identical blocks, each with S and volume V , along with their heat inputs Q .

4.2 Intensive parameters

Note we may write the total differential of U as a mathematical identity

$$dU = \left(\frac{\partial U}{\partial S} \right)_{V, \{X_i\}} dS + \left(\frac{\partial U}{\partial V} \right)_{S, \{X_i\}} dV + \sum_j \left(\frac{\partial U}{\partial X_j} \right)_{S, V, \{X_1, X_2, \dots, X_{j-1}, X_{j+1}, \dots\}} dX_j , \quad (4.2)$$

and using the first Law, i.e. Eq. (4.1). If we equate coefficients of the differentials we see:

$$\left(\frac{\partial U}{\partial S} \right)_{V, \{X_i\}} = T ; \quad \left(\frac{\partial U}{\partial V} \right)_{S, \{X_i\}} = -p ; \quad \left(\frac{\partial U}{\partial X_j} \right)_{S, V, \{X_1, X_2, \dots, X_{j-1}, X_{j+1}, \dots\}} = Y_j .$$

In principle one may solve for S as a function of $U, V, \{X_i\}$ and then form its total differential. It is however faster just to manipulate Eq. (4.1) to obtain:

$$dS = \frac{1}{T} dU + \frac{p}{T} dV + \sum_i \left(-\frac{Y_i}{T} \right) dX_i \quad (4.3)$$

where the entropic intensive parameters, sometimes called the *Massieu* functions.

Let us return to U . From the two equations

$$-p(S, V, \{X_i\}) = \left(\frac{\partial U}{\partial V} \right)_{S, \{X_i\}} \quad \text{and} \quad T(S, V, \{X_i\}) = \left(\frac{\partial U}{\partial S} \right)_{V, \{X_i\}} \quad (4.4)$$

we see in principle that could eliminate S from these two equations and end up with a relation between $p, T, V, \{X_i\}$, an example being the Equation of State for an ideal gas, $pV = nRT$. Sometimes the two equations Eq. (4.4) are termed the equations of state.

In fact $U(S, V, \{X_i\})$ (or $S(U, V, \{X_i\})$) contains all the thermodynamics information about the system.

4.3 Other thermodynamic potentials

The internal energy is a useful function when the entropy, S , is constant as in reversible adiabatic transformations, so the internal energy, $U(S, V, N)$ is useful. However often we work at constant temperature, T , or the pressure, p , is fixed rather than the volume, V . It is useful to work with functions which reflect the natural variables for the experiment.

We now list the choices for a gas. The gas is not necessarily an ideal gas, but still the state of the gas is described by (p, V) for example – not needing more variables.

We interchange within each pair of the intensive/extensive variables: T and S ; p and V (and, next year N and μ , the chemical potential).

This is achieved by a *Legendre transformation*, which appears magical, or a "sleight of hand" according to one author. There is a geometrical interpretation of this, e.g. [Making sense of the Legendre transform](#), R.K.P. Zia et al, *American Journal of Physics*, **77**, (2009), and chapter 10 of [Badger, Parker H., Equilibrium thermodynamics](#), Allyn and Bacon, (1967), but not very digestible.

- **Enthalpy, $H(S, p)$**

This is useful for adiabatic processes at constant pressure, we make the Legendre transform between V and p :

$$\begin{aligned} H &= U + pV \\ \Rightarrow dH &= dU + Vdp + pdV \end{aligned} \quad (4.5)$$

$$\begin{aligned} &= TdS - pdV + Vdp + pdV \\ &= TdS + Vdp . \end{aligned} \quad (4.6)$$

where we see from the total differential that $H(S, p)$, i.e. V has been replaced by p . We see if the coefficients in Eq. (4.6) are compared with

$$dH = \left(\frac{\partial H}{\partial S} \right)_p dS + \left(\frac{\partial H}{\partial p} \right)_S dp$$

we see: $\left(\frac{\partial H}{\partial S} \right)_p = T$ and $\left(\frac{\partial H}{\partial p} \right)_S = V$.

The Legendre transformation is invertible: $U = H - pV$, and all the thermodynamic information is also contained in it.

- **Helmholtz free energy, $F(T, V)$**

This is useful when the variables one controls (and may be fixed) are T and V .

$$F = U - TS$$

$$\Rightarrow dF = dU - SdT - TdS$$

(Use First law.) $= -SdT - pdV$

$$\Rightarrow S = - \left(\frac{\partial F}{\partial T} \right)_V \text{ and } p = - \left(\frac{\partial F}{\partial V} \right)_T$$

- **Gibbs free energy, $G(T, p)$**

This is useful with T and p are fixed or under control, we perform both Legendre transformations on U replacing S by T and V by p .

$$G = U - TS + pV$$

$$\Rightarrow dG = -SdT + Vdp$$

$$\Rightarrow \left(\frac{\partial G}{\partial T} \right)_p = -S \text{ and } \left(\frac{\partial G}{\partial p} \right)_V = V$$

4.4 Why is Free energy "free"?

Why is the Free energy termed so? Consider dF at constant T , i.e. an isothermal process:

$$dF = dU - TdS - \underbrace{SdT}_{=0}$$

(Use First Law $= dQ - dW_{by} - TdS$)

(Use Second Law) $\leq TdS - dW_{by} - TdS$

$$\Rightarrow dF \leq -dW_{by}$$

$$\Rightarrow -dF \geq dW_{by}$$
(4.7)

If the system is performing work the change in Free energy will be negative, so the magnitude of the Free energy change, $|dF| = -dF$, provides a bound on the work being done by the system, dW_{by} . It is in that sense that the Free energy is "free" - i.e. free to be turned into work, as against locked into heat which cannot be turned into work. Notice that the change in Free energy, ΔF , is smaller than the internal energy $\Delta U - T\Delta S$ (isothermal), showing that the magnitude of internal energy unrecoverable as work is $T\Delta S$.

4.5 Inequalities of thermodynamic potentials and evolution to equilibrium

Changes to the thermodynamic potentials obey inequalities:

$$\Delta S \geq 0 \quad \text{from Eq. (3.7)} \quad \Delta F \leq 0 \quad \text{from Eq. (4.7) and} \quad \Delta G \leq 0, \quad (4.8)$$

from analogous reasoning. These apply, respectively, when the system is isolated (U constant), the system has fixed temperature and volume, so no work performed, and at fixed temperature and pressure. So under those conditions, S is maximised, and F and G are minimised.¹

Let us consider how a system can relax to be in equilibrium with its surroundings, (T_0, p_0) , reversibly, performing work as this relaxation occurs. .

Imagine our system is initially at a higher temperature than its surroundings, $T > T_0$, and at a higher pressure than its surroundings, $p > p_0$. It could be regarded as a "resource", from which we wish to extract work as it relaxes, in a reversible (quasi-static) manner to T_0, p_0 .

The system can do work by expanding into surroundings, until the pressures equalise, $p = p_0$. And by using a subsidiary engine running a Carnot cycle (to maximise the amount of work performed), with the body playing the role of the high-temperature reservoir work is performed and some heat deposited in the surroundings at T_0 , playing the role of the low-temperature bath.

4.6 Maxwell relations

Not all quantities are equally easy to measure. For T we have a thermometer from the zeroth law, for V we have a metre stick (if the system is a cuboid), for pressure, p , we may measure with a spring. But entropy is not straightforward, and its derivatives even less so.

There is a trick to hand: Maxwell relations. These are examples of the rule that for a smooth-enough function of two variables, $f(x, y)$ (see Maths 1B), then

$$\frac{\partial^2 f}{\partial x \partial y} = \frac{\partial^2 f}{\partial y \partial x}. \quad (4.9)$$

¹It is surprisingly tricky to establish in general that these inequalities are related to the second derivatives of the thermodynamic potentials – as one would normally find easily when considering stationary points of a function of two variables, (U, V) , (T, V) and (T, p) respectively. See [A. H. Wilson, *Thermodynamics and Statistical Mechanics*, CUP \(1957\)](#), p. 56-61.

We may apply this to the relations in Eq. (4.4) we see

$$\begin{aligned} \left[\frac{\partial}{\partial V} \left(\frac{\partial U}{\partial S} \right)_V \right]_S &= \left[\frac{\partial}{\partial S} \left(\frac{\partial U}{\partial V} \right)_S \right]_V \\ \Rightarrow \left(\frac{\partial T}{\partial V} \right)_S &= - \left(\frac{\partial p}{\partial S} \right)_V, \end{aligned} \quad (4.10)$$

Note the inverses are equal (see Eq. (A.6))

$$\left(\frac{\partial V}{\partial T} \right)_S = - \left(\frac{\partial S}{\partial p} \right)_V,$$

so we may relate the RHS, which involves entropy and is hard to measure to the LHS which needs a thermometer and a metre rule.

We may derive counterparts from the other thermodynamic potentials:

$$\text{(From enthalpy:)} \quad \left(\frac{\partial p}{\partial T} \right)_V = \left(\frac{\partial S}{\partial V} \right)_T \quad (4.11)$$

$$\text{(From Helmholtz:)} \quad \left(\frac{\partial V}{\partial S} \right)_p = \left(\frac{\partial T}{\partial p} \right)_S \quad (4.12)$$

$$\text{(From Gibbs:)} \quad \left(\frac{\partial S}{\partial p} \right)_T = - \left(\frac{\partial V}{\partial T} \right)_p. \quad (4.13)$$

This provides most of the derivatives of S apart from $(\partial S / \partial T)_{p \text{ or } V}$, which we will find in the final subsection. Some other partial derivatives may be related by the identity (see Eq. (A.9))

$$\left(\frac{\partial z}{\partial x} \right)_y \left(\frac{\partial x}{\partial y} \right)_z \left(\frac{\partial y}{\partial z} \right)_x = -1,$$

used in problem sheet 3.

4.7 Heat capacities

so far we have defined the heat capacities indirectly via

$$\begin{aligned} \mathrm{d}Q &= C_V \mathrm{d}T && \text{at constant volume} \\ \mathrm{d}Q &= C_p \mathrm{d}T && \text{at constant pressure} \end{aligned}$$

These expressions, whilst often convenient, do not provide a direct definition of C_V or C_p .

Firstly let us deal with constant volume. There $\mathrm{d}W = 0$ if work is performed by a piston, say. Thus by the First law, $\mathrm{d}U = \mathrm{d}Q$, and hence $\mathrm{d}Q$ is a perfect differential, $\mathrm{d}Q = C_V \mathrm{d}T$.

Thus

$$\begin{aligned}
 dU &= C_V dT \\
 \Rightarrow C_V &= \left(\frac{\partial U}{\partial T} \right)_V \\
 \text{(But since } dS &= \frac{dU}{T} + \frac{p}{T} \underbrace{dV}_{=0} = \frac{C_V}{T} dT \\
 \Rightarrow C_V &= T \left(\frac{\partial S}{\partial T} \right)_V .
 \end{aligned}$$

For constant pressure, it is natural to use the enthalpy, H .

$$\begin{aligned}
 H &= U + pV \\
 \Rightarrow dH &= dU + d(pV) \\
 &= dQ - p dV + p dV + V dp \\
 &= dQ + V \underbrace{dp}_{=0} \\
 &= C_p dT \\
 \Rightarrow \left(\frac{\partial H}{\partial T} \right)_p &= C_p \\
 \text{(Then, since:)} \quad dS &= \frac{dH}{T} - \frac{V}{T} \underbrace{dp}_{=0} \\
 \Rightarrow \left(\frac{\partial S}{\partial T} \right)_p dT &= \frac{1}{T} \left(\frac{\partial H}{\partial T} \right)_p dT \\
 \Rightarrow T \left(\frac{\partial S}{\partial T} \right)_p &= C_p .
 \end{aligned}$$

Appendices

A Results on partial derivatives from Maths 1B

A.1 Implicit differentiation

Suppose we define a relation between x and y implicitly via $f(x, y) = 0$. In simple cases such as

$$f(x, y) = x + y - 1 = 0 \quad \Rightarrow \quad y = 1 - x ,$$

we may solve explicitly for $y(x)$. But usually not, eg if there is a mixture of powers and trigonometric or exponential functions, for example:

$$f(x, y) = x \cos y + y \cos x - 1 = 0 . \quad (\text{A.1})$$

This cannot be solved for either x or y .

How can we calculate dy/dx ?

Consider the total differential, df , which vanishes as $f(x, y) = 0$:

$$\begin{aligned} df = 0 &= \left(\frac{\partial f}{\partial x} \right)_y dx + \left(\frac{\partial f}{\partial y} \right)_x dy \\ \Rightarrow \frac{dy}{dx} &= - \frac{(\partial f / \partial x)_y}{(\partial f / \partial y)_x} . \end{aligned} \quad (\text{A.2})$$

We denote which variable is being kept constant by the subscript of the brackets enclosing the derivative. In this case it is obvious, but as we see later this is not always so clear (example 4).

Note that naïve Leibnitz does not work in Eq. (A.2) – one cannot "cancel the ∂f 's", as there is a minus sign. This is because the variables which are being kept constant are different in the numerator and denominator.

Example 1 – mixture of trigonometric functions and powers

We consider Eq. (A.1)

$$\begin{aligned} f(x, y) &= x \cos y + y \cos x - 1 = 0 \\ \Rightarrow \left. \begin{aligned} \left(\frac{\partial f}{\partial x} \right)_y &= \cos y - y \sin x \\ \left(\frac{\partial f}{\partial y} \right)_x &= -x \sin y + \cos x \end{aligned} \right\} &\Rightarrow \frac{dy}{dx} = - \frac{(\partial f / \partial x)_y}{(\partial f / \partial y)_x} = - \frac{\cos y - y \sin x}{\cos x - x \sin y} . \end{aligned}$$

Example 2 – an ellipse

If one can uncover a geometric picture, it can help understanding the problem.

What is dy/dx at the point $(x, y) = (1, 1)$, where $f(x, y)$ is defined by

$$f(x, y) = x^2 + xy + y^2 - 3 = 0 ? \quad \Rightarrow \quad \frac{3}{4}(x + y)^2 + \frac{1}{4}(x - y)^2 = 3 ,$$

This is an ellipse with its axes at $\pi/4$ to the x - and y -axes. We should expect that dy/dx will

hence diverge at two points.

$$\begin{aligned}
 f(x, y) &= x^2 + xy + y^2 - 3 = 0 \\
 \Rightarrow \left\{ \begin{aligned} \left(\frac{\partial f}{\partial x} \right)_y &= 2x + y \\ \left(\frac{\partial f}{\partial y} \right)_x &= 2y + x \end{aligned} \right\} &\Rightarrow \frac{dy}{dx} = -\frac{(\partial f / \partial x)_y}{(\partial f / \partial y)_x} = -\frac{2x + y}{2y + x} \\
 \Rightarrow \left. \frac{dy}{dx} \right|_{(1,1)} &= -\frac{3}{3} = -1.
 \end{aligned}$$

Note that the derivative does diverge at the two points where the line $2y + x = 0$ intersects the ellipse. Upon substitution for $x = -2y$ into $f(x, y) = 0$, we find the points are $(x, y) = (\pm 2, \mp 1)$.

A.2 Three or more dimensions

The procedure is more complicated with three variables – as in the thermodynamics for a gas.

Consider the function:

$$\begin{aligned}
 f(x, y, z) &= 0 \Rightarrow z = z(x, y) \\
 \Rightarrow df = 0 &= \left(\frac{\partial f}{\partial x} \right)_{yz} dx + \left(\frac{\partial f}{\partial y} \right)_{zx} dy + \left(\frac{\partial f}{\partial z} \right)_{xy} dz \\
 &= f_x dx + f_y dy + f_z dz
 \end{aligned} \tag{A.3}$$

We also have, since $z(x, y)$,

$$dz = \left(\frac{\partial z}{\partial x} \right)_y dx + \left(\frac{\partial z}{\partial y} \right)_x dy. \tag{A.4}$$

We wish to find these partial derivatives (which are the counterpart to dy/dx in the two variable case).

The problem is that we cannot now simply divide the differentials in Eq. (A.3) to obtain a counterpart to dy/dx in last subsection. We need to select the variable whose partial derivatives we want – $z(x, y)$ here – and then we replace dz in Eq. (A.3) using Eq. (A.4). Then

$$\begin{aligned}
 df = 0 &= \left(\frac{\partial f}{\partial x} \right)_{yz} dx + \left(\frac{\partial f}{\partial y} \right)_{zx} dy + \left(\frac{\partial f}{\partial z} \right)_{xy} \left[\left(\frac{\partial z}{\partial x} \right)_y dx + \left(\frac{\partial z}{\partial y} \right)_x dy \right] \\
 \Rightarrow 0 &= \left[\left(\frac{\partial f}{\partial x} \right)_{yz} + \left(\frac{\partial f}{\partial z} \right)_{xy} \left(\frac{\partial z}{\partial x} \right)_y \right] dx + \left[\left(\frac{\partial f}{\partial y} \right)_{zx} + \left(\frac{\partial f}{\partial z} \right)_{xy} \left(\frac{\partial z}{\partial y} \right)_x \right] dy
 \end{aligned}$$

So we choose one variable to keep fixed, say $dy = 0$, then the coefficient of dx must vanish so:

$$\begin{aligned} 0 &= \left(\frac{\partial f}{\partial x} \right)_{yz} + \left(\frac{\partial f}{\partial z} \right)_{xy} \left(\frac{\partial z}{\partial x} \right)_y \\ \Rightarrow \left(\frac{\partial z}{\partial x} \right)_y &= - \frac{(\partial f / \partial x)_{yz}}{(\partial f / \partial z)_{xy}} = - \frac{f_x}{f_z} \end{aligned} \quad (\text{A.5})$$

and note that y is, consistently, being kept constant in all three partial derivatives.

Note the above result implies (by interchanging x and y) that

$$\left(\frac{\partial x}{\partial y} \right)_z = \left[\left(\frac{\partial y}{\partial x} \right)_z \right]^{-1} \quad (\text{A.6})$$

Similarly, we see for the other two partial derivatives:

$$\left(\frac{\partial x}{\partial y} \right)_z = - \frac{f_y}{f_x} \quad \text{and} \quad \left(\frac{\partial y}{\partial z} \right)_x = - \frac{f_z}{f_y}. \quad (\text{A.7})$$

We may combine Eq. (A.5) and Eq. (A.7) by multiplying the three LHS together and similarly the three RHS:

$$\left(\frac{\partial z}{\partial x} \right)_y \left(\frac{\partial x}{\partial y} \right)_z \left(\frac{\partial y}{\partial z} \right)_x = (-1)^3 \frac{f_x f_y f_z}{f_z f_x f_y} \quad (\text{A.8})$$

$$= -1 \quad (\text{A.9})$$

A disconcerting result – the -1 .

Example 3 – a three dimensional case

Consider an example where one cannot solve for one of the variables, z in this case:

$$f(x, y, z) = z + \sin(xyz) - 1 = 0$$

$$\Rightarrow \left\{ \begin{array}{l} f_x = yz \cos(xyz) \\ f_y = zx \cos(xyz) \\ f_z = 1 + xy \cos(xyz) \end{array} \right\} \Rightarrow \left\{ \begin{array}{l} \left(\frac{\partial z}{\partial x} \right)_y = - \frac{\partial f / \partial x}{\partial f / \partial z} = - \frac{yz \cos(xyz)}{1 + xy \cos(xyz)} \\ \left(\frac{\partial z}{\partial y} \right)_x = - \frac{\partial f / \partial y}{\partial f / \partial z} = - \frac{zx \cos(xyz)}{1 + xy \cos(xyz)} \end{array} \right.$$

Example 4 – the importance of what is constant

The following illustrates the possibility of contradictions if one is not careful about recognising which variable (or variables) are being kept constant.

Here, as usual, x and y are Cartesian coordinates and r and θ are polar coordinates related by

$$\begin{aligned} x &= r \cos \theta & y &= r \sin \theta \\ r &= \sqrt{x^2 + y^2} & \tan \theta &= y/x \end{aligned}$$

Is

$$\frac{\partial \theta}{\partial x} \stackrel{?}{=} \left(\frac{\partial x}{\partial \theta} \right)^{-1} ?$$

Let us evaluate both sides of this equation:

$$\begin{aligned} \theta = \arctan(y/x) & \Rightarrow \frac{\partial \theta}{\partial x} = \frac{1}{1 + (y/x)^2} \left(-\frac{y}{x^2} \right) = -\frac{y}{r^2} \\ x = r \cos \theta & \Rightarrow \frac{\partial x}{\partial \theta} = -r \sin \theta = -y \end{aligned}$$

Thus

$$\frac{\partial \theta}{\partial x} = -\frac{y}{r^2} \neq \left(\frac{\partial x}{\partial \theta} \right)^{-1} = -\frac{1}{y}$$

The reason for the disparity is not being careful about which variables are being kept constant.

On the left it is y and on the right it is r . To evaluate $(\partial x / \partial \theta)_y$, express x in terms of y and θ : $x = r \cos \theta$ but $y = r \sin \theta$ so $r = y / \sin \theta$ and $x = y \cot \theta$. thus:

$$\left(\frac{\partial x}{\partial \theta} \right)_y = -\frac{y}{\sin^2 \theta} = -\frac{r}{\sin \theta} \Rightarrow \left[\left(\frac{\partial x}{\partial \theta} \right)_y \right]^{-1} = -\frac{\sin \theta}{r} = \frac{\partial \theta}{\partial x} \text{ as above.}$$

Example 5 – the ideal gas

For an ideal gas, the equation of state is: $pV = nRT$. So we may define a function:

$$f(p, V, T) = 0 = pV - nRT, \quad (\text{A.10})$$

which is simple enough that we may solve for any of the three variables:

$$p(V, T) = \frac{nRT}{V} \quad (\text{A.11}) \quad V(p, T) = \frac{nRT}{p} \quad (\text{A.12}) \quad T(p, V) = \frac{pV}{nR} \quad (\text{A.13})$$

and construct the partial derivatives directly. We will use both approaches, checking they agree.

Method 1 For example

$$\text{From Eq. (A.11): } \left(\frac{\partial p}{\partial V} \right)_T = -\frac{nRT}{V^2}, \text{ From Eq. (A.12): } \left(\frac{\partial V}{\partial p} \right)_T = -\frac{nRT}{p^2},$$

and note reassuringly

$$\left(\frac{\partial p}{\partial V} \right)_T \left(\frac{\partial V}{\partial p} \right)_T = (-1)^2 \frac{nRT}{V^2} \frac{nRT}{p^2} = \left(\frac{nRT}{pV} \right)^2 = 1,$$

using the equation of state.

Method 2 Use a general equation of state, $f(p, V, T) = 0$, i.e. not necessarily an ideal gas

Then

$$\left(\frac{\partial p}{\partial V}\right)_T = -\frac{\partial f/\partial V}{\partial f/\partial p} \quad \text{and} \quad \left(\frac{\partial V}{\partial p}\right)_T = -\frac{\partial f/\partial p}{\partial f/\partial V}.$$

So:

$$\left(\frac{\partial p}{\partial V}\right)_T \left(\frac{\partial V}{\partial p}\right)_T = (-1)^2 \frac{\partial f/\partial V}{\partial f/\partial p} \frac{\partial f/\partial p}{\partial f/\partial V} = 1.$$

The same result as before.

More surprisingly, noting which variables are constant,

$$\left(\frac{\partial p}{\partial V}\right)_T \left(\frac{\partial V}{\partial T}\right)_p \left(\frac{\partial T}{\partial p}\right)_V = \left(-\frac{\partial f/\partial V}{\partial f/\partial p}\right) \left(-\frac{\partial f/\partial T}{\partial f/\partial V}\right) \left(-\frac{\partial f/\partial p}{\partial f/\partial T}\right) = (-1)^3 = -1.$$

Note the sign!

Example 6 – the van der Waals equation of state

Start with van der Waals Equation of state:

$$f(p, V, T) = \frac{nRT}{V - nb} - \frac{n^2a}{V} - p = 0, \quad (\text{A.14})$$

Then the implicit procedure readily provides, for example:

$$\left(\frac{\partial V}{\partial T}\right)_p = -\frac{\partial f/\partial T}{\partial f/\partial V} = \frac{\frac{-nR}{(V - nb)}}{-\frac{2n^2a}{V^3} + \frac{nRT}{(V - nb)^2}} = -\frac{nR(V - nb)V^3}{nRTV^3 - 2n^2a(V - nb)^2}.$$

For the virial expansion (to which the Van der Waals equation of state is an approximation), see Problem 4.10 problem sheet 4.